



Minor Project 2

Objective

Each student is required to independently complete a minor project based on a real-world problem using an **Unsupervised Machine Learning** approach. The project should demonstrate the complete machine learning workflow, from problem identification to model evaluation and reporting.

Project Requirements

1. Problem Selection

- Identify a relevant real-world problem that can be solved using unsupervised machine learning.(for example : customer segmentation)
- Clearly define the problem statement and project objective.

2. Dataset Selection

- Find a suitable dataset from reliable sources such as Kaggle, UCI Repository, Government Open Data portals, etc.
- The dataset must be relevant to the chosen problem.
- Provide the dataset source/link in the report.

3. Data Preprocessing

Perform appropriate preprocessing steps, which may include:

- Handling missing values
- Removing duplicates
- Feature encoding
- Feature scaling/normalization
- Exploratory Data Analysis (EDA)



4. Model Development

- Train at least one unsupervised machine learning model.
- Suitable models may include:
 - Kmeans
 - DBSCAN
 - Heirarichal Clustering
 - Any other unsupervised learning algorithm

5. Model Evaluation

Use appropriate evaluation metrics based on the problem type:

Classification:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Provide proper analysis and interpretation of the results.



6. Project Report

Submit a report containing:

1. Title
2. Problem Statement
3. Dataset Description
4. Data Preprocessing Steps
5. Exploratory Data Analysis
6. Model Used
7. Training Process
8. Evaluation Metrics and Results
9. Conclusion
10. References

ANSWER THE FOLLOWING AND UPLOAD IT IN THE REPORT...

Semantic vs Instance vs Panoptic Segmentation.

What is **Fuzzy Logic**, and how is it different from Boolean Logic?

Suppose I give you a customer dataset containing **Age, Annual Income, and Spending Score**, but I don't provide any labels.

1. Is this a supervised or unsupervised learning problem?
2. Which clustering algorithm would you choose first and why?
3. What assumptions does that algorithm make about the data?

Submission Requirements

Submit:

- Source Code (Jupyter Notebook / Python File)
- Project Report (PDF Format)
- Dataset Link

Important Instructions

- The project must be completed individually.
- Plagiarism or copied work will result in disciplinary action.
- Proper documentation and explanation of each step are mandatory.
- Projects without evaluation and analysis will not be considered complete.



Deadline

5 July 2026, 5:00 PM

Late submissions will not be accepted.